

Midsem 22 Solutions.

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FLSMIDTH

- 1) Assume that $m(t)$ is a real baseband signal with bandwidth w , and given signals

$$x_p(t) = m(t) \cos 2\pi f_0 t$$

$$y_p(t) = m(t) \sin 2\pi f_0 t, \quad f_0 > w$$

Compute inner products in passband and in the complex baseband, i.e.,

Compute $\langle x_p, y_p \rangle$ and $\langle x, y \rangle$, where

$x(t)$ and $y(t)$ are complex baseband representation of $x_p(t)$ & $y_p(t)$, respectively.

Solution:

x_I components: $x_I(t) = m(t)$

$$y_I(t) = 0.$$

Q - components: $x_Q(t) = 0$

$$y_Q(t) = -m(t)$$

$$\langle x_p, y_p \rangle = \frac{1}{2} (\langle x_I, y_I \rangle + \langle x_Q, y_Q \rangle)$$

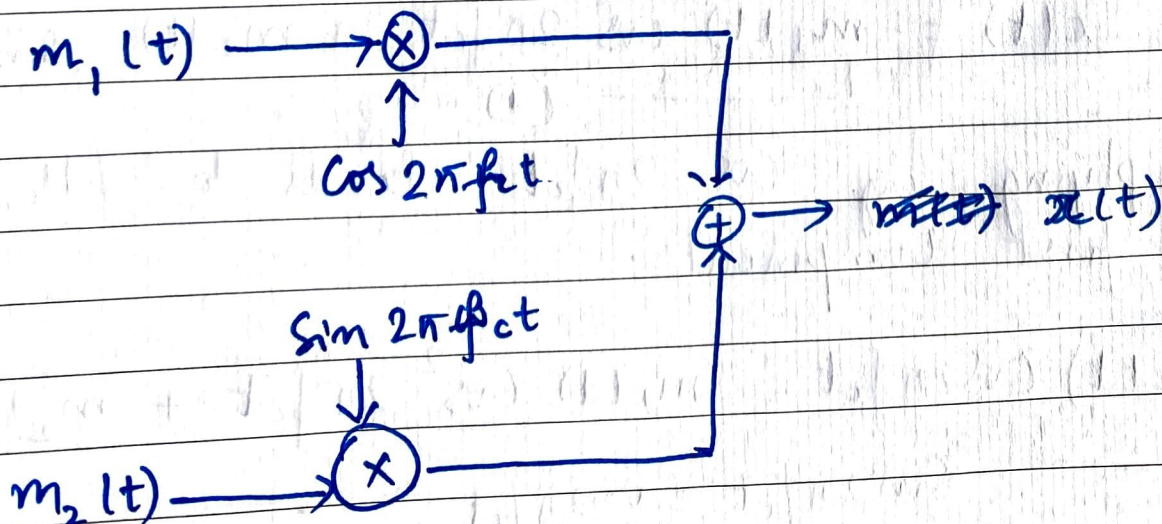
$$= \frac{1}{2} (\langle m, 0 \rangle + \langle 0, -m \rangle)$$

$$= 0. \quad \text{--- (2)}$$

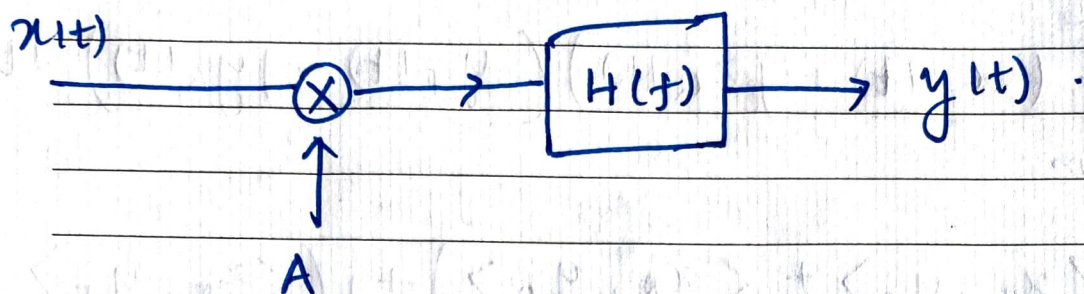
$$\begin{aligned}
 \langle x, y \rangle &= \int_{-\infty}^{\infty} x(t) y^*(t) dt \\
 &= \int_{-\infty}^{\infty} (x_I(t) + j x_Q(t)) (y_I(t) - j y_Q(t)) dt \\
 &= (\langle x_I, y_I \rangle + \langle x_Q, y_Q \rangle) + j (\langle x_Q, y_I \rangle - \langle x_I, y_Q \rangle) \\
 &= 0 + j (\langle 0, 0 \rangle - \langle m(t), -m(t) \rangle) \\
 &= j \langle m(t), m(t) \rangle \quad \text{--- (3)}
 \end{aligned}$$

⇒ Orthogonality in passband does not imply orthogonality in Complex Baseband. --- (1)

2) Consider a transmitter as given below.



Let the demodulator be of the form:



Provide expressions for A & ~~Sketch~~ plot $H(f)$ if:

* ~~y(t)~~ We need to obtain $y(t) = \frac{1}{2} m_1(t)$.

* We need to obtain $y(t) = m_2(t)$.

Solution:

(a) For $x(t) = m_1(t) \cos 2\pi f_c t + m_2(t) \sin 2\pi f_c t$ (1)

Multiplying by $\cos 2\pi f_c t$ leads to filtering the first component.

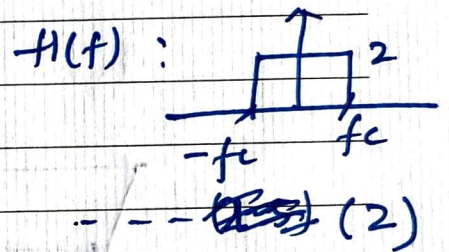
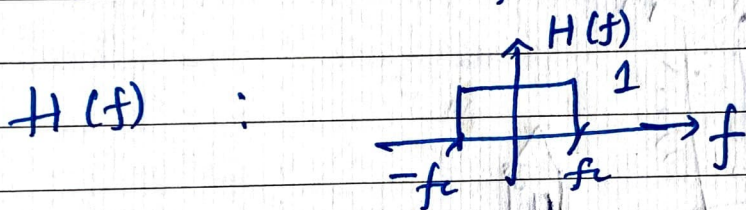
$$\Rightarrow x(t) \cos 2\pi f_c t = m_1(t) \cos^2 2\pi f_c t + m_2(t) \sin 2\pi f_c t \cos 2\pi f_c t$$

- - - (1)

$$\Rightarrow x(t) \cos 2\pi f_c t = \frac{m_1(t)}{2} + \frac{m_1(t) \cos 4\pi f_c t}{2} + \frac{m_2(t)}{2} \sin 4\pi f_c t$$

$\Rightarrow$ If $H(f)$ has cut-off at $-f_c$ & f_c , only $m_1(t)/2$ is filtered.

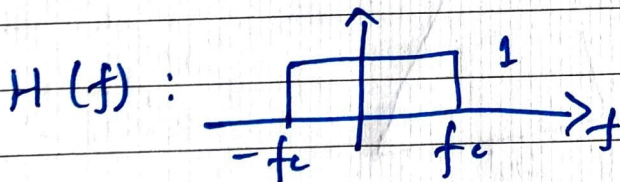
$$\Rightarrow A = 2 \cos 2\pi f_c t \quad \text{or} \quad A: \cos 2\pi f_c t$$



(b) Similarly ~~for~~ $y(t) = m_2(t)$

$$A = 2 \sin 2\pi f_c t$$

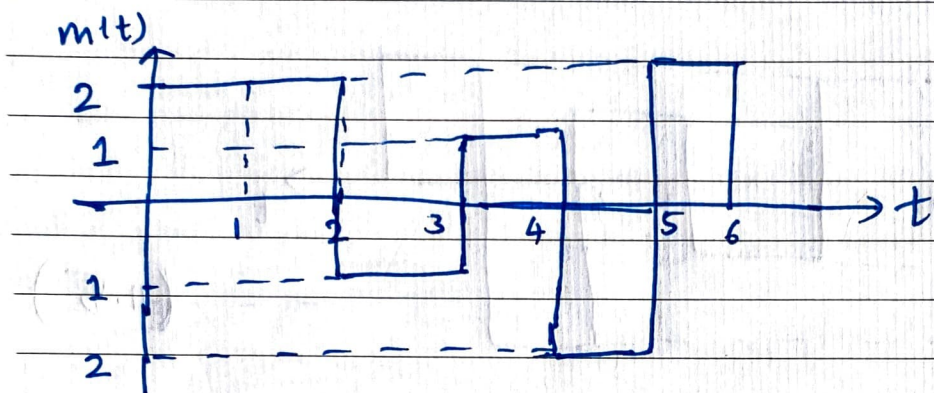
or //



(Marks distribution similar to last time. All steps need to be written. Students cannot write the 'similarly' argument)

--- (3)

3) Consider an FM signal with $k_f = 20\pi$.
The message signal is given by:



(a) Plot the frequency deviation of the FM signal as a function of time.

~~(b) Plot the phase deviation of the FM signal as a function of time.~~

Solution: It is given that it is an FM wave.

⇒ Instantaneous freq (angular):

$$\omega_i(t) = \omega_c + k_f m(t).$$

$$\Rightarrow f_i(t) = f_c + \frac{k_f}{2\pi} m(t) \quad \text{--- (1)}$$

⇒ Instantaneous freq must be deviation:

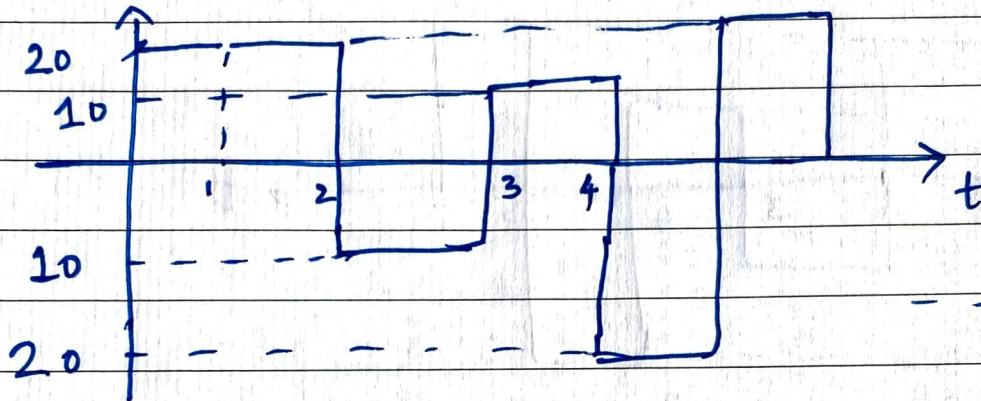
$$f_i(t) - f_c = \frac{k_f}{2\pi} m(t)$$

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Given $k_f = 20\pi \Rightarrow \frac{k_f}{2\pi} = 10 \dots (0.5)$

~~(2)~~

hence:



~~(11.5)~~

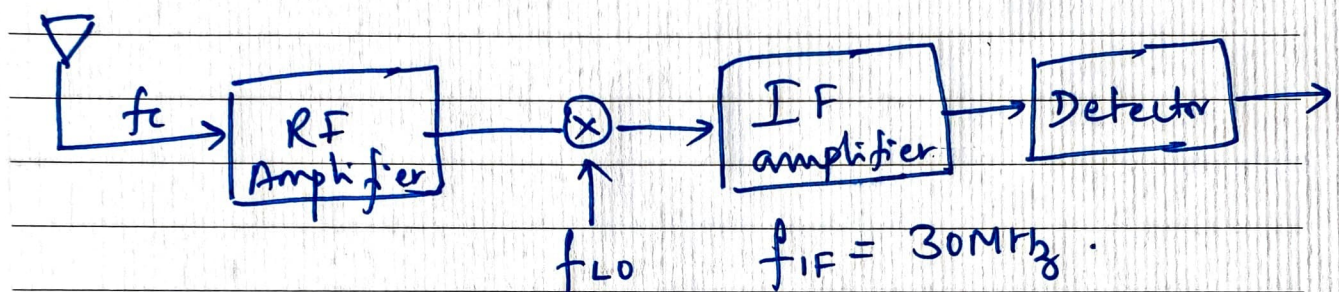
$$\int_0^2 2 \, dt$$

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4.) The telemetry receiver shown below is designed so that local oscillator operates at freq below incoming carrier frequencies.



(a) What is f_{LO} set to when 136 MHz Carrier signal is received.

Ans: $106 \text{ MHz} = f_c - f_{IF} \text{ --- (1)}$

(b) What is image frequency for this Carrier frequency? Ans: 76 MHz --- (2)
($f_c - 2f_{IF}$)

c) If receiver is designed to receive signals from 120 MHz to 150 MHz , what tuning range is required for the local oscillator?

Ans: $90 - 120 \text{ MHz}$ --- (2)